Online Resource 1 – Supplementary Information

International Journal of Applied and Computational Mathematics

Sample-Efficient Residual-Adaptive Collocation for Physics-Informed Neural Networks: A 270-Run Study of Shuttlecock Flipping Dynamics

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S1. Experiment integrity and checkpoint-stage selection

The final dataset contains 270 unique method–budget–seed runs (3 methods × 3 budgets × 30 paired seeds), with no missing or duplicate run keys. Figure 1 summarizes whether the final accepted model was the best Adam checkpoint or the post-L-BFGS checkpoint selected by the independent physics-validation criterion.

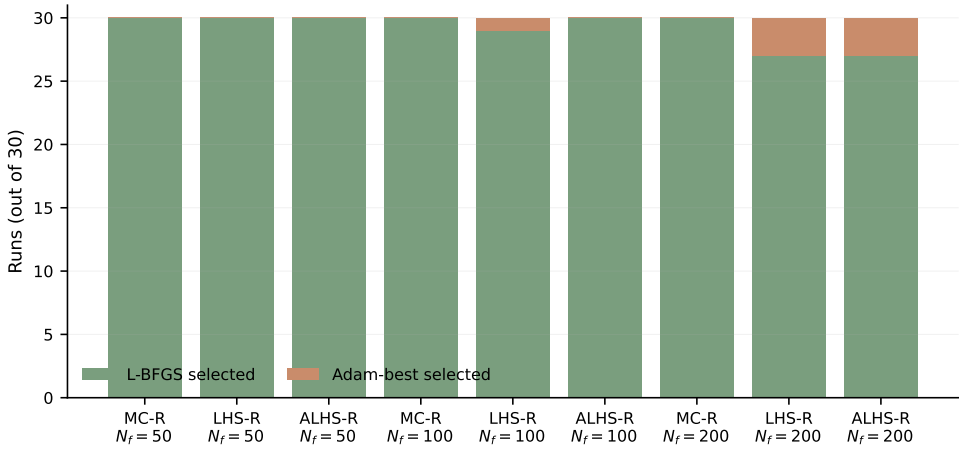


Figure 1: Final checkpoint-stage selection across 30 seeds for each method and collocation budget.

S2. Omnibus method comparisons

Table 1: Friedman tests comparing MC-R, LHS-R, and ALHS-R at fixed collocation budgets. Kendall's W is the omnibus effect size.

Metric	N_f	χ^2_F	p	Kendall W
ϵ_{ϕ,L_2}	50	23.267	8.866e-06	0.388
ϵ_{ϕ,L_2}	100	35.267	2.198e-08	0.588
ϵ_{ϕ,L_2}	200	18.200	1.117e-04	0.303
ϵ_{ω,L_2}	50	24.867	3.984e-06	0.414
ϵ_{ω,L_2}	100	33.800	4.575e-08	0.563
ϵ_{ω,L_2}	200	16.267	2.936e-04	0.271
$L_{\text{phys,dense}}$	50	2.067	3.558e-01	0.034
$L_{\text{phys,dense}}$	100	1.267	5.308e-01	0.021
$L_{\text{phys,dense}}$	200	1.800	4.066e-01	0.030
Runtime	50	32.600	8.337e-08	0.543
Runtime	100	42.467	6.005e-10	0.708
Runtime	200	6.467	3.943e-02	0.108

S3. Pairwise method comparisons for trajectory error

Table 2: Holm-corrected paired Wilcoxon signed-rank comparisons. Positive matched rank-biserial r_{rb} indicates that method B has lower error than method A.

Metric / N_f	Comparison	p_{Holm}	r_{rb}	Median A/B
Angle / 50	MC-R vs LHS-R	5.769e-02	-0.398	0.48
Angle / 50	MC-R vs ALHS-R	6.899e-04	0.712	3.31
Angle / 50	LHS-R vs ALHS-R	1.157e-06	0.927	6.93
Angle / 100	MC-R vs LHS-R	6.906e-05	-0.802	0.27
Angle / 100	MC-R vs ALHS-R	1.529e-04	0.746	3.52
Angle / 100	LHS-R vs ALHS-R	1.062e-07	0.970	12.82
Angle / 200	MC-R vs LHS-R	9.515e-01	0.015	1.01
Angle / 200	MC-R vs ALHS-R	1.333e-03	0.682	2.76
Angle / 200	LHS-R vs ALHS-R	7.689e-04	0.725	2.73
Angular velocity / 50	MC-R vs LHS-R	6.356e-02	-0.389	0.56
Angular velocity / 50	MC-R vs ALHS-R	2.748e-04	0.751	2.88
Angular velocity / 50	LHS-R vs ALHS-R	1.414e-06	0.923	5.13
Angular velocity / 100	MC-R vs LHS-R	6.906e-05	-0.802	0.29
Angular velocity / 100	MC-R vs ALHS-R	4.601e-04	0.699	2.73
Angular velocity / 100	LHS-R vs ALHS-R	1.062e-07	0.970	9.36
Angular velocity / 200	MC-R vs LHS-R	9.515e-01	0.015	1.01
Angular velocity / 200	MC-R vs ALHS-R	1.596e-03	0.673	2.62
Angular velocity / 200	LHS-R vs ALHS-R	1.140e-03	0.708	2.60

S4. Collocation-budget effects

Table 3: Friedman tests across $N_f = 50, 100, 200$ within each method.

Metric	Method	χ^2_F	p	Kendall W
ϵ_{ϕ,L_2}	MC-R	7.200	2.732e-02	0.120
ϵ_{ϕ,L_2}	LHS-R	4.867	8.774e-02	0.081
ϵ_{ϕ,L_2}	ALHS-R	18.067	1.194e-04	0.301
ϵ_{ω,L_2}	MC-R	7.267	2.643e-02	0.121
ϵ_{ω,L_2}	LHS-R	4.867	8.774e-02	0.081
ϵ_{ω,L_2}	ALHS-R	20.067	4.391e-05	0.334

Table 4: Selected paired cross-budget comparisons illustrating ALHS-R sample efficiency for relative angle error. Negative r_{rb} means configuration A has lower error than configuration B. Holm correction is applied within the eight pre-specified cross-budget comparisons for this metric.

Configuration A	Configuration B	p_{Holm}	r_{rb}
ALHS-R N=50	MC-R N=100	9.149e-05	-0.806
ALHS-R N=50	LHS-R N=100	1.490e-08	-1.000
ALHS-R N=50	MC-R N=200	4.657e-08	-0.987
ALHS-R N=50	LHS-R N=200	1.490e-08	-1.000
ALHS-R N=100	MC-R N=200	4.657e-08	-0.987
ALHS-R N=100	LHS-R N=200	1.490e-08	-1.000

S5. Final checkpoint counts

Table 5: Final checkpoint stage selected by independent physics-validation loss.

N_f	Method	Adam best	L-BFGS
50	MC-R	0	30
50	LHS-R	0	30
50	ALHS-R	0	30
100	MC-R	0	30
100	LHS-R	1	29
100	ALHS-R	0	30
200	MC-R	0	30
200	LHS-R	3	27
200	ALHS-R	3	27